

Mid - Agile Management Exam Prep

Agile Management & Prioritization:

- The Requirements Hierarchy (Business, Functional, Non-Functional)
- The MoSCoW Prioritization Method
- Core Scrum Concepts (Roles and Events related to Backlog Management)

Teach me everything you know about these with examples.
Make me fully prepared for exam

Absolutely! I'll break down each topic in detail, with real-world examples, step-by-step explanations, and practical exam-focused notes. Let's make sure you're **exam ready** for all three: Requirements Hierarchy, MoSCoW Prioritization, and Core Scrum Concepts.

1. Requirements Hierarchy

What is it?

The requirements hierarchy is a way of organizing requirements in levels according to their scope and detail. It helps teams understand which requirements are most important for the business, how they break down into system features, and what constraints or qualities (like security or usability) are needed.

Levels of Requirements

1. Business Requirements

- *Definition:* High-level needs of the organization or goals for the project.
- *Example:* "Increase online sales by 20% in one year."
- *How to spot them:* They answer "Why are we doing this project?"

2. Functional Requirements

- *Definition:* Specific features, functions, or behaviors the system must have to achieve business requirements.

- *Example:* "The system shall allow users to pay with credit card or mobile wallet."
- *How to spot them:* They describe **what** the system must do.

3. Non-Functional Requirements

- *Definition:* Quality attributes, constraints, or conditions the system must meet.
 - *Examples:*
 - Performance: "System must process transactions within 3 seconds."
 - Security: "User data must be encrypted in transit and at rest."
 - Usability: "The app should be usable by a first-time user with no training."
 - *How to spot them:* They describe **how well** the system works or under what conditions.
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Visual Analogy

Imagine building a house:

- **Business Requirement:** "We need a place to live" (the big goal)
 - **Functional Requirement:** "House must have 3 bedrooms, 2 bathrooms, a kitchen, etc." (the features)
 - **Non-Functional Requirement:** "The house should be energy efficient, safe in earthquakes, easy to clean, etc." (the qualities/standards)
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Example Scenario

Suppose a company wants to launch a **food delivery app**.

- **Business Requirement:** "Expand to a new market and capture 10,000 users in 6 months."
 - **Functional Requirement:** "Users can track delivery status in real time."
 - **Non-Functional Requirement:** "App uptime must be 99.9% per month."
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2. MoSCoW Prioritization Method

What is it?

A popular method to prioritize requirements and features based on their necessity for the project's success.

MoSCoW stands for:

- **Must have**
- **Should have**
- **Could have**
- **Won't have (this time)**

How does it work?

You sort all requirements into one of these four categories. This helps teams and stakeholders agree on what's truly essential and what can be delayed if necessary.

Definitions & Examples

- **Must Have:** Absolutely essential. Project fails without these.
 - *Example:* "Login functionality for users."
- **Should Have:** Important but not vital; there's a workaround.
 - *Example:* "Password reset via email." (Can still use manual support as a backup)
- **Could Have:** Nice to have; adds value if included.
 - *Example:* "Dark mode for the app."
- **Won't Have:** Not doing it now; maybe later.
 - *Example:* "Integration with smart fridge" (future version)

Practical Example (E-commerce Site)

Requirement	MoSCoW Category
Checkout & payment system	Must Have
Product reviews	Should Have

Requirement	MoSCoW Category
Wishlist feature	Could Have
Virtual try-on for clothes	Won't Have

Why is MoSCoW useful?

- Prevents scope creep.
 - Ensures the minimum viable product (MVP) can be delivered on time.
 - Aligns stakeholders on priorities.
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3. Core Scrum Concepts (Roles and Events related to Backlog Management)

Scrum in a Nutshell

Scrum is an Agile framework used to develop, deliver, and sustain complex products. It's built around roles, events, and artifacts. **Backlog management** is key in Scrum—it's how you organize and prioritize what to build next.

Key Scrum Roles

1. Product Owner (PO)

- *Responsibilities:* Owns the Product Backlog; decides what the team should build next based on value, ROI, and stakeholder needs.
- *Example action:* Adds new features and prioritizes them.

2. Scrum Master

- *Responsibilities:* Facilitates the Scrum process, removes impediments, coaches the team.
- *Example action:* Helps the team have efficient meetings.

3. Development Team

- *Responsibilities:* Builds and delivers increments of the product each Sprint.
 - *Example action:* Estimates backlog items and commits to Sprint Goals.
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Key Scrum Events (related to Backlog Management)

1. Sprint Planning

- *Purpose:* Decide what work will be done in the upcoming Sprint.
- *Action:* The team selects items from the Product Backlog to move into the Sprint Backlog.

2. Daily Scrum (Standup)

- *Purpose:* Short daily meeting to synchronize activities.
- *Action:* Team discusses progress, blockers, and next steps.

3. Sprint Review

- *Purpose:* Show completed work to stakeholders; get feedback.
- *Action:* Demonstrate new features, discuss what went well and what didn't.

4. Sprint Retrospective

- *Purpose:* Reflect and improve team processes.
 - *Action:* Team identifies what to keep, improve, or stop doing.
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Product Backlog

- **Definition:** An ordered list of everything that might be needed in the product.
- **Managed by:** Product Owner.
- **Contents:** Features, bug fixes, technical work, knowledge acquisition.

Sprint Backlog

- **Definition:** The set of Product Backlog items selected for the Sprint, plus a plan for delivering them.

- **Managed by:** Development Team.
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Example (Backlog Item Lifecycle)

1. **Product Owner** adds a new requirement: “Add PayPal support.”
 2. **Backlog Refinement:** Team discusses it, estimates effort, clarifies details.
 3. **Sprint Planning:** Team selects “Add PayPal support” for next Sprint.
 4. **Sprint Review:** Demonstrate PayPal payment feature to stakeholders.
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Exam Tips & How to Answer Questions

- Define each concept in your own words, not just memorized lines.
 - Always give an example—real or imaginary.
 - If given a scenario, show how to apply MoSCoW or break down requirements hierarchy.
 - When asked about Scrum, describe roles AND events, and how backlog management fits in.
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Quick Practice Questions

Q1: Break down this requirement: “The system should allow customers to order food online quickly and safely.”

- **Business Requirement:** Increase food orders via online system.
 - **Functional Requirement:** Customers can place food orders.
 - **Non-Functional Requirement:** Orders must be processed within 2 minutes and with secure payment.
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Q2: Classify the following using MoSCoW:

- “User login” – Must Have
 - “Admin analytics dashboard” – Should Have
 - “User avatar customization” – Could Have
 - “Integration with VR headset” – Won’t Have
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Q3: Who owns the Product Backlog in Scrum?

- **Answer:** Product Owner.
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Q4: What is the purpose of Sprint Planning?

- **Answer:** To decide what work will be accomplished in the next Sprint, moving items from the Product Backlog to the Sprint Backlog.
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Summary Table

Topic	Definition	Key Example	Why Important
Requirements Hierarchy	Organize requirements: Business → Functional → Non-Functional	E-commerce: “Increase sales” → “Cart”	Ensures alignment & clarity
MoSCoW Prioritization	Must, Should, Could, Won’t	“Login” = Must, “Wishlist” = Could	Sets clear priorities
Scrum Roles & Events	PO, Scrum Master, Dev Team + Sprint events	PO prioritizes; team delivers features	Organizes teamwork, ensures focus

Last-Minute Revision: Key Points

- **Hierarchy:** Business (why) → Functional (what) → Non-Functional (how well/under what conditions).
- **MoSCoW:** Always sort requirements—know which are *essential* vs *nice-to-have*.
- **Scrum:** PO manages backlog; team commits via Sprint Planning; feedback via Sprint Review; process improvement via Retrospective.

If you want **more sample questions/answers or case scenarios**, just ask!

Ready for any last-minute drills or deep-dives!

explain MoSCoW Prioritization Method with more details

Absolutely! Here's a **detailed explanation** of the **MoSCoW Prioritization Method**—with practical steps, examples, tips for exams, and real-world use.

What is MoSCoW Prioritization?

MoSCoW is a powerful method used in Agile, project management, software development, and business analysis to clearly prioritize features, requirements, or tasks.

- It helps teams and stakeholders agree on what is truly essential for a project and what can be delayed or dropped if time/resources run short.

What Does MoSCoW Stand For?

The acronym breaks down as:

Letter	Meaning	What it means for your project
M	Must Have	Essential requirements. The project cannot succeed without these.
S	Should Have	Important, but not critical. Workarounds exist; not delivery-stoppers.

Letter	Meaning	What it means for your project
C	Could Have	Desirable if time/resources permit; lower impact if left out.
W	Won't Have (this time)	Not being included now; maybe considered for the future, but out of scope.

How to Apply MoSCoW Prioritization (Step by Step):

1. List all requirements/features/tasks.
2. Discuss with stakeholders/team.
3. Categorize each item as Must, Should, Could, or Won't have.
4. Revisit and adjust categories as needed during the project (priorities may shift).
5. Use the categories to plan releases, sprints, or project phases.

In-Depth: What Each Category Means

1. Must Have

- **Definition:** Absolutely critical for launch/success.
- If even one Must Have is missing, the product fails or is not viable.
- **Examples:**
 - For an online banking app: Secure login, ability to view account balance.
 - For an e-commerce site: Add to cart, checkout, payment processing.

2. Should Have

- **Definition:** Important, but not vital for initial launch; workarounds are possible.
- If Should Haves are left out, the product is less usable or less attractive, but still functional.
- **Examples:**

- Forgot password/reset option (could use manual customer support as a temporary workaround).
- Filter by price in search results (can still search, but less user-friendly).

3. Could Have

- **Definition:** Nice to have, adds value or convenience, but not critical.
- **Usually low-cost enhancements or “icing on the cake.”**
- **Examples:**
 - Dark mode theme.
 - Social media sharing buttons.
 - Custom avatars for users.

4. Won't Have (this time)

- **Definition:** Not planned for this release/phase, out of scope. May revisit later.
- **Helps prevent scope creep and keeps the team focused.**
- **Examples:**
 - Voice command ordering in a food delivery app.
 - Integration with wearable devices.

MoSCoW in Practice: Example Table

Example: Food Delivery App

Requirement	MoSCoW Category
User registration/login	Must Have
Browse restaurants	Must Have
Place an order	Must Have
Track order in real time	Should Have
Pay with Apple Pay/Google Pay	Should Have

Requirement	MoSCoW Category
Save favorite orders	Could Have
Ratings and reviews	Could Have
AR-based menu preview	Won't Have

Benefits of MoSCoW

- **Clarity:** Everyone knows what's truly critical.
- **Focus:** Avoids wasting effort on low-impact features.
- **Negotiation:** Makes it easier to negotiate scope with stakeholders.
- **Agility:** When time or resources are limited, focus on Musts first, then Shoulds, and so on.

MoSCoW in Agile Projects

- Used heavily in **Sprint Planning** or **Release Planning**.
- Product Owner works with stakeholders to assign priorities.
- If the deadline is tight, Coulds and some Shoulds may be dropped to ensure delivery.

Tips for Exam Answers

- **Define the method:** "MoSCoW is a prioritization technique that categorizes requirements as Must, Should, Could, and Won't have, helping teams deliver the most critical elements first."
- **Give an example:** Use a relatable product (e.g., banking app, e-commerce site).
- **Explain why it's valuable:** Mention scope management, delivery focus, and stakeholder alignment.

Common Mistakes/What NOT To Do

- Don't make everything a Must Have! (Defeats the purpose.)
- Don't forget to revisit priorities—business needs can change.
- Don't use "Won't Have" for things that are secretly essential; it's for true "not now."

Key MoSCoW Exam Question Example

Q: "You're developing an e-commerce website. How would you use the MoSCoW method to prioritize the following features: Search, Wishlist, Guest Checkout, Integration with AR, Social Sharing, Payment Processing?"

Sample Answer:

Using MoSCoW prioritization, I would classify 'Search' and 'Payment Processing' as Must Have, because the website cannot function without them. 'Guest Checkout' and 'Wishlist' would be Should Have, as they improve the user experience but are not strictly required for basic operation. 'Social Sharing' could be categorized as Could Have, adding value if time permits. 'Integration with AR' would be Won't Have (this time), as it's an advanced feature outside the current project scope.

Summary Table

MoSCoW Category	Importance	Example Feature	Impact if Missing
Must Have	Critical	Secure login	Project fails
Should Have	Important	Forgot password	Workaround possible
Could Have	Nice to have	Dark mode	No major impact
Won't Have	Not for now	Voice commands	No impact, may add later

If you want, I can give you more case studies, sample questions, or a MoSCoW exercise for practice!

give me case study

Of course! Here's a **full MoSCoW case study** you can use to practice and understand how to apply the method in a real-world scenario. This will help you answer exam questions and handle practical prioritization tasks in Agile environments.

MoSCoW Case Study: Online Learning Platform

Background:

A startup is building an online learning platform for university students. The project must be delivered in **12 weeks** for the upcoming semester. The team has limited developers and budget. Key stakeholders include university administrators, teachers, and students.

List of Requirements:

1. Student registration and login
2. Video lectures upload and streaming
3. Course enrollment by students
4. Assignment submission and grading
5. Real-time chat between students and teachers
6. Push notifications for deadlines
7. User profile picture upload
8. Multi-language support (English, Bangla, Hindi)
9. Analytics dashboard for teachers
10. Integration with Zoom for live classes
11. Dark mode option
12. Mobile app version

Step 1: Stakeholder Discussion

The Product Owner gathers input from all stakeholders about what's absolutely needed for a successful launch.

Step 2: Applying MoSCoW

Must Have

- **Student registration and login** (Students can't use the system without it)
- **Video lectures upload and streaming** (Core content delivery method)
- **Course enrollment by students** (Basic need for access)
- **Assignment submission and grading** (Key function for course completion)

Should Have

- **Real-time chat between students and teachers** (Enhances interaction; if missing, could use email as a backup)
- **Push notifications for deadlines** (Improves engagement, but students can use calendar reminders if missing)
- **Analytics dashboard for teachers** (Helpful for teachers, but basic usage reports can be emailed manually at first)

Could Have

- **User profile picture upload** (Nice for personalization, but not necessary)
- **Dark mode option** (Enhances user comfort, not essential)
- **Multi-language support (English, Bangla, Hindi)** (Could reach more users, but can launch with English only first)

Won't Have (this time)

- **Integration with Zoom for live classes** (Zoom links can be shared manually; integration can be added in the next release)
- **Mobile app version** (Focus on web platform for initial launch due to time/resource constraints)

Final MoSCoW Table:

Requirement	MoSCoW Category
Student registration and login	Must Have
Video lectures upload and streaming	Must Have
Course enrollment by students	Must Have
Assignment submission and grading	Must Have
Real-time chat	Should Have
Push notifications	Should Have
Analytics dashboard	Should Have
Profile picture upload	Could Have
Dark mode	Could Have
Multi-language support	Could Have
Zoom integration	Won't Have
Mobile app version	Won't Have

How this helps the project:

- Ensures all critical learning features are ready for semester start.
 - If the team is running out of time, they'll focus on Must Haves first, then Should Haves if possible.
 - Could Haves and Won't Haves are postponed, so the team isn't overwhelmed or distracted.
 - Stakeholders are aligned and know what to expect in the first release.
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Sample Exam-Style Question:

Q: Given the requirements above and a tight 12-week timeline, show how you would use MoSCoW prioritization for the online learning platform. Explain why you categorized each item as you did.

How to Answer:

I used MoSCoW prioritization to categorize the requirements for the online learning platform. I placed core functionalities like student registration, course enrollment, video lectures, and assignment handling as Must Have because they are essential for the system's operation. Features like chat, notifications, and analytics are categorized as Should Have since they add value but can be temporarily replaced by manual workarounds. Could Have features such as profile pictures and dark mode improve user experience but are not critical. Features like Zoom integration and a mobile app are classified as Won't Have this time, as they require significant effort and can be added in future versions. This approach ensures we deliver a usable product on time and manage stakeholder expectations.

If you want, I can give you another case study in a different domain (e.g., e-commerce, healthcare, banking) or walk you through this one step by step. Let me know!

Explain the requirement hierarchy in more details please

Absolutely! Let's dive deep into the **Requirements Hierarchy**—what it is, why it matters, how to distinguish the levels, and how it looks in real projects.

What is Requirements Hierarchy?

The **Requirements Hierarchy** is a structured way of organizing requirements from the most general, high-level business needs to the most detailed technical specifications. It helps

ensure that what is built directly supports the organization's goals and meets users' expectations.

The Three Main Levels

1. Business Requirements
2. Stakeholder & User Requirements
3. System/Functional & Non-Functional Requirements

(Some models split Stakeholder/User as a separate level, but most Agile and exam settings focus on Business → Functional → Non-Functional.)

1. Business Requirements

- **Definition:** High-level goals or objectives that the organization wants to achieve.
- **Ask:** "Why are we doing this project?"
- **Who sets them?** Senior management, sponsors, business owners.

Examples:

- "Increase market share by 15% in the next year."
- "Comply with new government data privacy regulations."
- "Reduce order processing time to under 2 hours."

Exam Point:

Business requirements describe **the value the business seeks** and are usually broad, strategic, and measurable.

2. Functional Requirements

- **Definition:** Specific capabilities, features, or functions the system must provide to meet the business goals.

- **Ask:** "What should the system do?"
- **Who defines them?** Business analysts, product owners, development teams (based on business and stakeholder needs).

Examples:

- "System shall allow users to search for products by name or category."
- "Users can upload documents to their personal profile."
- "Admin can generate monthly sales reports."

Key Characteristics:

- Describes **actions, processes, or rules** the system must perform.
- Directly supports business requirements.

3. Non-Functional Requirements (NFRs)

- **Definition:** The qualities, constraints, or standards the system must meet. Not about **what** it does, but **how** it does it.
- **Ask:** "How well should the system perform?" "What standards must it meet?"
- **Who cares?** End-users, system architects, regulatory bodies.

Common Types and Examples:

NFR Type	Example
Performance	"System must process 1,000 transactions per second."
Security	"All data must be encrypted at rest and in transit."
Usability	"First-time users must be able to use the app with no training."
Reliability	"System must have 99.95% uptime."
Compliance	"System must follow GDPR standards."

How Do These Levels Connect?

Think of it as a funnel:

- **Business Requirement:** "Reduce customer complaints by 40%."
 - Leads to...
 - **Functional Requirement:** "Implement a customer feedback form on the website."
 - Supported by...
 - **Non-Functional Requirement:** "The feedback form must load within 2 seconds and be available 24/7."

Another Example:

Level	Requirement Example
Business	"Expand service to new cities."
Functional	"Allow users to select their city during registration."
Non-Functional	"Support must be available in both English and Bengali."

Visual Analogy

- **Building a Car:**
 - **Business Requirement:** "Develop an eco-friendly vehicle for urban commuters."
 - **Functional Requirement:** "Car must operate using electric power."
 - **Non-Functional Requirement:** "Car must travel at least 300 km on a single charge and meet safety standards."

Why is Requirements Hierarchy Important?

- **Ensures alignment:** Every feature and function supports a business goal.
- **Prevents confusion:** Clear separation of what, why, and how.

- **Helps traceability:** If a functional requirement is challenged, you can link it back to the business need.

Hierarchical Structure Table

Level	Key Question	Example
Business	Why?	"Increase customer satisfaction."
Functional	What?	"Provide real-time order tracking."
Non-Functional	How well/under what?	"Updates must display in under 5 sec."

How to Write or Identify Each Type (Exam Tips):

- **Business Requirements:** Start with business outcomes ("increase," "reduce," "achieve," "comply").
- **Functional Requirements:** Use "The system/app shall/should..." and describe features, workflows, data handling.
- **Non-Functional Requirements:** Use words about quality, performance, security, compliance, "within," "no longer than," "must be able to."

Sample Scenario:

E-commerce App

Hierarchy Level	Example Requirement
Business	"Increase completed purchases by 20% in 6 months."
Functional	"Enable users to save and reuse shipping addresses."
Non-Functional	"Saved addresses must be encrypted and load within 1 second."

Advanced: Stakeholder/User Requirements (optional for some exams)

Sometimes, you may see:

- **Stakeholder/User Requirements:** Needs/wants of specific users or groups.
 - *Example:* "As a customer, I want to view my order history."

These often act as a bridge between business and functional requirements.

Summary Table

Level	Focus	Typical Wording/Question	Example
Business Requirement	Goals/Strategy	Why do this?	"Reduce costs by 10%."
Functional Requirement	Features/Actions	What must system do?	"Allow users to track deliveries online."
Non-Functional Req.	Quality/Standards	How well/under what limits?	"Deliveries must be tracked in real time."

Key Points for Exam:

- Always relate lower-level requirements to higher-level business goals.
 - Non-functional requirements are just as important—don't ignore them.
 - Give examples for each level.
 - Use clear wording to distinguish each level.
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Want more sample questions or scenario-based exercises? Let me know!

Absolutely! Here's a **deep-dive guide** to the **core Scrum concepts**, focusing on the roles and events that are essential for backlog management and Agile success. This will cover everything you need for your exam and practical understanding.

Scrum Concepts Explained in Detail

1. Scrum Roles

Scrum defines **three key roles**—each with unique responsibilities, especially around the **Product Backlog** (the list of everything to be done).

A. Product Owner (PO)

- **Responsibilities:**
 - Owns and manages the **Product Backlog**.
 - Decides what the team works on next based on business value and stakeholder input.
 - Sets priorities, accepts or rejects work, and ensures requirements are clear.
- **Backlog Management:**
 - Adds new items, clarifies details, prioritizes features using techniques like MoSCoW.
 - Regularly refines and reorders the backlog as needs change.
- **Exam Example:**
 - *If a stakeholder wants a new feature, the PO decides where it fits in the backlog based on importance and impact.*

B. Scrum Master

- **Responsibilities:**

- Acts as a **coach and facilitator**.
- Ensures the Scrum process is followed.
- Removes obstacles that block the team's progress.
- Facilitates Scrum events (meetings) and helps the team self-organize.
- **Backlog Management:**
 - Supports the Product Owner and team in backlog refinement.
 - Helps resolve misunderstandings or blockers about backlog items.
- **Exam Example:**
 - *If a developer doesn't understand a backlog item, the Scrum Master ensures the PO provides clarification.*

C. Development Team

- **Responsibilities:**
 - Self-organizing group (usually 3-9 members).
 - Builds the product incrementally in Sprints.
 - Selects which items from the Product Backlog to work on in each Sprint (creates the Sprint Backlog).
 - Collaborates to estimate, design, build, and test features.
 - **Backlog Management:**
 - Estimates the effort for each backlog item.
 - Commits to items they can deliver within the Sprint.
 - **Exam Example:**
 - *If the team cannot complete all selected items, they update the Sprint Backlog and inform the PO.*
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2. Scrum Artifacts (Related to Backlog)

- **Product Backlog:**
 - A dynamic, ordered list of everything the product might need.

- Managed by the Product Owner.
 - Can include features, bug fixes, tech improvements, research, etc.
 - **Sprint Backlog:**
 - The subset of the Product Backlog chosen for the current Sprint (2–4 weeks).
 - Owned and managed by the Development Team.
 - Contains all items the team commits to delivering during the Sprint, plus a plan.
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3. Scrum Events (Ceremonies) and Backlog Management

A. Sprint

- A time-boxed period (typically 2–4 weeks) where the team works to deliver a usable increment.
- No changes allowed to Sprint Backlog during the Sprint (except by team agreement).

B. Sprint Planning

- **Purpose:**
 - Decides **what** will be delivered in the coming Sprint and **how** it will be done.
- **Process:**
 - Product Owner presents prioritized Product Backlog items.
 - Team discusses, clarifies, and selects which items to move into the Sprint Backlog.
 - Team breaks selected items into actionable tasks.
- **Exam Example:**
 - *If the PO's highest-priority item is too big, the team splits it into smaller parts for the Sprint.*

C. Daily Scrum (Daily Standup)

- **Purpose:**
 - A 15-minute meeting for the Development Team to coordinate and share progress.
- **Focus:**

- “What did I do yesterday?”
- “What will I do today?”
- “Are there any blockers?”
- **Relation to Backlog:**
 - Progress on Sprint Backlog items is discussed.
 - Problems with backlog items can be raised for quick resolution.

D. Sprint Review

- **Purpose:**
 - Demonstrate completed work to stakeholders.
 - Gather feedback on the Increment (the potentially shippable product).
- **Process:**
 - The team presents finished Sprint Backlog items.
 - Stakeholders suggest changes or new backlog items.
 - Product Owner updates Product Backlog based on feedback.
- **Exam Example:**
 - *If feedback reveals a missing feature, the PO adds it to the Product Backlog.*

E. Sprint Retrospective

- **Purpose:**
 - Reflect as a team on how the Sprint went and how to improve.
- **Focus:**
 - What went well?
 - What could be improved?
 - What will we try next Sprint?
- **Relation to Backlog:**
 - Process improvements might lead to new backlog items (e.g., automate testing, refactor code).

F. Backlog Refinement (Grooming)

- **Purpose:**
 - Ongoing activity (not an official event) where the team and PO review and improve backlog items.
- **Activities:**
 - Clarify requirements, split large items, re-estimate, remove outdated items.
- **Benefits:**
 - Keeps the backlog “ready” for future Sprints.

Example Workflow

1. **Product Owner** adds “Payment by mobile wallet” to the Product Backlog.
2. **Backlog Refinement:** Team estimates effort, discusses acceptance criteria.
3. **Sprint Planning:** Team selects the item for the next Sprint.
4. **Daily Scrum:** Team tracks progress and discusses any issues.
5. **Sprint Review:** Demo the mobile wallet payment feature.
6. **Sprint Retrospective:** Team discusses what worked/needs improvement, maybe adds a “refactor payment module” item to the backlog.

Summary Table: Scrum Roles and Events

Role/Artifact/Event	Main Responsibility	Exam Example
Product Owner	Owns Product Backlog, sets priorities	Adds/removes items, clarifies requirements
Scrum Master	Coaches team, ensures Scrum is followed	Removes blockers, facilitates meetings
Development Team	Builds increment, selects Sprint Backlog	Estimates work, delivers features
Product Backlog	List of all work, owned by PO	Continuously refined and prioritized

Role/Artifact/Event	Main Responsibility	Exam Example
Sprint Backlog	Committed Sprint work, team-owned	Created in Sprint Planning
Sprint Planning	Decide Sprint goals and backlog	Team and PO select & plan items
Daily Scrum	Sync team, resolve blockers	Discuss Sprint Backlog progress
Sprint Review	Demo, gather feedback, update backlog	Stakeholders suggest new items
Sprint Retrospective	Process improvement	Identify actions for next Sprint
Backlog Refinement	Make backlog “ready” for future Sprints	Split/clarify/prioritize items

Key Points for Exams

- Always **define each role and event** and connect them to backlog management.
- Give **practical examples** (see above).
- Mention that **backlog refinement** is a continuous, collaborative process.
- Emphasize that the **Product Owner owns the Product Backlog**, but the whole team is involved in refinement and estimation.

Quick Practice Questions

1. Who is responsible for prioritizing the Product Backlog?
 - *Answer:* Product Owner.
2. What is the purpose of Sprint Planning?
 - *Answer:* To decide what work will be accomplished in the next Sprint and how to do it.
3. What happens in a Sprint Review?

- *Answer:* The team demonstrates completed work, gets feedback, and updates the Product Backlog.

4. Why is backlog refinement important?

- *Answer:* It keeps backlog items clear, appropriately sized, and prioritized for upcoming Sprints.
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Let me know if you want **sample scenarios, more exam questions, or diagrams** for these concepts!